



DOMAIN 2 : SMART AGRIBUSINESS & CLIMATE DEFENCE FOR CITRUS

TECH STACK

INTRODUCTION

Agriculture plays an important role in the economy and livelihoods of communities in the Lowveld region. The region has favourable conditions for producing valuable crops such as macadamia nuts and citrus fruits, which contribute significantly to local and international agricultural markets. Many farmers depend on reliable crop production for their income, employment opportunities and business growth.





PROBLEM STATEMENT

However :

- climate variability is creating challenges for farmers because conditions such as soil moisture and temperature can change significantly.
- One major problem is insufficient soil moisture. When there is not enough water in the soil, crops can experience water stress, which can negatively affect their growth and production. On the other hand, excessive irrigation can waste valuable water and increase farming costs.
- Another problem is high temperatures. Extreme heat can cause stress to plants and may affect crop development and quality. Farmers therefore need to know when and where conditions are becoming dangerous so that they can take action quick
- It can be difficult and time-consuming to manually check every area. Problems may only be noticed after the crops have already been affected. This creates a need for a system that can continuously monitor environmental conditions and identify stressed areas automatically.



SOLUTION:

- The proposed solution is a smart agricultural monitoring and climate defence system that uses environmental sensors to monitor conditions in macadamia and citrus farms.
- Senses soil moisture and temperature.
- A wind speed that will detect strong wind that can damage crops.
- Buzzer that will give alert severe crops stress.
- WIFI will allow the 3D digital twin to receive live data.
- The device will detect how fast the soil is drying and how fast the temperature is rising.
- Estimate hours until the critical stress to the plant(Predictive stress code).
- Prioritise sections to know which sides to get water first.

